

SOFTWARE REQUIREMENTS SPECIFICATION

Project Name

Course Registration and Recommendation System

Group -11

Prepared by

Roll	Name	Role
202480090226	Numaier Islam Khan	Leader
202480090104	MD Tasnimul Hasan	Team member
202480090108	Shahedul Alam	Team member
202480090184	MD Zayed Shikder	Team member
202480090200	MD Fahamud Alam Chowdhury Fuad	Team member
202480090222	Abdullah Al Toha	Team member

Course: Software Requirements Engineering

Department of Software Engineering

ZHENGZHOU UNIVERSITY

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional and non-functional requirements for version 1.0 of the University Course Registration & Recommendation System. The purpose of this document is to establish a clear, shared understanding of what the system is expected to do and the constraints under which it must operate. It serves as the primary reference point for the development team, who will design and build the system; for the testers, who will verify that the delivered product meets the specified requirements; and for the project stakeholders, including university administrators and academic advisors, who need to confirm that the system aligns with institutional policies and academic workflows. By documenting both what the system shall do and what it shall not do, this SRS aims to reduce ambiguity, prevent scope creep, and provide a measurable baseline against which the final product can be validated.

1.2 Document Conventions

To ensure consistency and clarity throughout this document, the following conventions are used:

- "Shall" indicates a mandatory requirement. If a feature is described with "shall," it must be implemented for the system to be considered compliant.
- "Should" indicates a recommended requirement. These are important but may be deferred if necessary, provided the reason is documented.
- "May" indicates an optional feature that could be included in a future release.
- Priority levels are assigned to each requirement using the following scale:
    - [Must] — Compulsory; the system cannot be accepted without it.
    - [Should] — Important; strongly desirable and expected in the final product.
    - [Could] — Nice to have; may be included if time and resources permit.

All requirements are uniquely identified (e.g., FR-1, NFR-1) so that they can be traced throughout the design, implementation, and testing phases.

1.3 Project Scope

The system will allow university students to securely register for courses, view academic histories, and use a unified smart wallet for campus payments. It will also provide academic advisors with tools to approve registrations and administrators with the ability to manage courses, users, and registration periods.

Out of Scope: Attendance tracking, Health & Support Services (these will be handled by a separate Learning Management

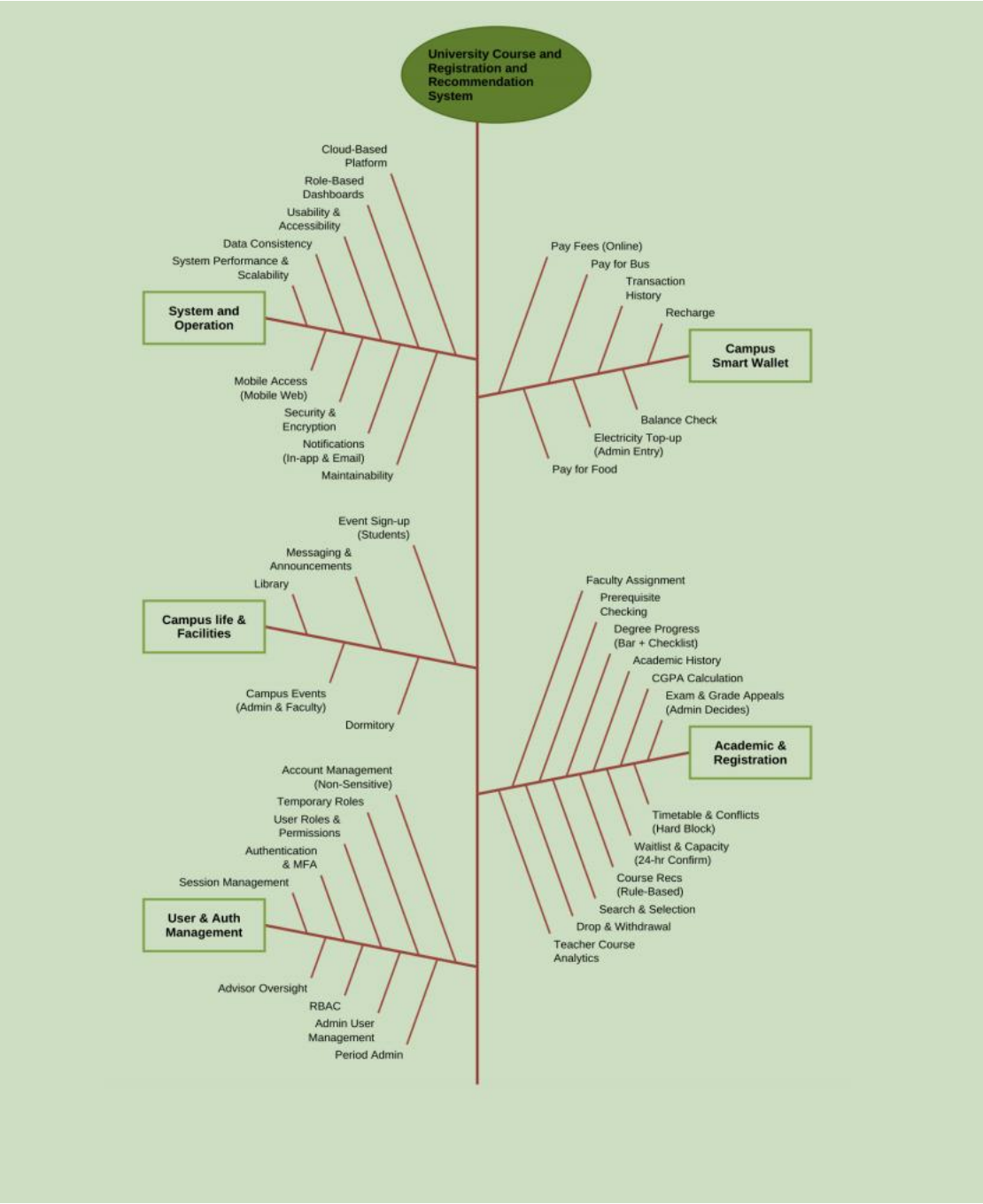


Figure 1.3: Feature Tree of University Course Registration and Recommendation System

1.4 References

1. IEEE Std 830-1998, *IEEE Recommended Practice for Software Requirements Specifications*, IEEE Computer Society, 1998.
2. ISO/IEC/IEEE 29148:2018, *Systems and software engineering — Life cycle processes — Requirements engineering*, International Organization for Standardization, 2018.
3. Week-01 to Week-05 Team Meeting Minutes and Initial/Updated Requirement Lists.
4. Lecture 8: Software Requirements Specification II.

2. Overall Description

2.1 Product Perspective

The system replaces the traditional manual course registration process. It integrates course management, registration, a recommendation engine, and a campus smart wallet into a single web and mobile-accessible platform.

2.2 User Classes and Characteristics

USER CLASS	DESCRIPTION	CHARACTERISTICS
Student:	Students can log in to the system, view available courses, register for courses, check registration status, view academic results, check their class timetable, monitor attendance, and use the Smart Wallet to manage tuition and other campus-related payments.	High frequency of use. Students frequently access the system to manage their academic activities, course registration, results, and payments.
Teacher:	Teachers can view their assigned courses, manage course-related information, access student lists, upload and update grades, monitor student attendance, and communicate important academic information to students.	Medium frequency of use. Teachers mainly use the system during teaching activities, attendance management, and examination or grading periods.
Advisor:	Advisors can review student course registration requests, approve or reject registrations, check students' academic records, monitor academic progress, identify academic problems, and provide guidance regarding course selection and academic planning.	Medium frequency of use. Advisors mainly use the system during course registration periods and when students require academic guidance or approval.
Administrator:	Administrators have overall control of the system. They can manage user accounts and permissions, maintain course information, configure system settings, manage academic semesters, monitor system activities, and maintain important institutional data.	High frequency of use. Administrators regularly access the system to manage users, update academic information, maintain system operations, and ensure proper system functionality.

Table 2.2 User Classes and Characteristics

2.3 Operating Environment

OE-1: The system shall operate correctly on modern web browsers, including Google Chrome, Mozilla Firefox, Microsoft Edge, and Apple Safari (current and latest versions).

OE-2: The system shall operate on a secure cloud server environment (e.g., AWS or Azure) running a standard web server (e.g., Apache or Nginx) and a secure database system (e.g., MySQL or PostgreSQL).

OE-3: The system shall permit user access from both desktop devices and mobile devices (Android and iOS) via responsive web browsers or mobile applications.

OE-4: The system shall be able to communicate with external environments, including third-party payment gateways (for the Smart Wallet) and the university's existing Single Sign-On (SSO) system.

2.4 Design and Implementation Constraints

CO-1: The system shall be developed using a secure and modern web framework (e.g., React/Node.js, Python/Django, or Java/Spring Boot) to ensure high performance and maintainability.

CO-2: The system shall use a relational database management system (e.g., MySQL or PostgreSQL) to ensure data consistency for course registrations and wallet transactions.

CO-3: The system shall implement strict Role-Based Access Control (RBAC) to ensure that students, teachers, advisors, and admins can only access the features permitted by their roles.

CO-4: The system shall use secure API integrations (e.g., HTTPS/TLS) for communicating with external third-party payment gateways for the Smart Wallet.

CO-5: The system's user interface shall be designed to be responsive, ensuring compatibility with both desktop and mobile screen sizes without requiring separate codebases**.**

2.5 Assumptions and Dependencies

Assumptions:

AS-1: It is assumed that the university administration will provide the master list of courses, credit hours, and prerequisite rules before the registration period begins.

AS-2: It is assumed that all students and faculty members have access to a stable internet connection and a valid university ID to log into the system.

AS-3: It is assumed that the university's academic calendar (registration dates, drop deadlines) is finalized before the system is deployed for that term.

Dependencies:

DE-1: The successful operation of the Smart Wallet depends on external third-party payment gateways (e.g., Stripe, PayPal, or local bank APIs) to process online top-ups.

DE-2: The authentication system depends on the university's existing Single Sign-On (SSO) service (e.g., Google Workspace or Azure AD) for secure user login.

DE-3: The course recommendation engine depends on accurate historical academic data (past grades and completed courses) being migrated into the new database**.**

3. System Features

3.1 User & Auth Management

3.1.1 Description

This feature handles secure user login, session management, and strict role-based access control. It ensures that only authorized users can access specific system features (e.g., Admins managing courses, Students registering for classes).

Priority = High.

Requirement Name	Details
User Login & MFA	The system shall require Multi-Factor Authentication (MFA) for Teachers, Advisors, and Administrators. Students shall log in using the University's Single Sign-On (SSO).
Session Management	The system shall automatically log out a user after 15 minutes of inactivity. Users shall also have the option to securely log out manually.
Account Lockout	The system shall temporarily lock a user account after 5 consecutive failed login attempts to prevent unauthorized access.
Role-Based Access Control (RBAC)	The system shall restrict user access based on their roles (Student, Teacher, Advisor, Admin). A student shall not be able to access administrative dashboards.
Admin User Management	The Administrator shall be able to create new user accounts, update user information, deactivate accounts, and assign specific roles to users.
Advisor Assignment	The Administrator shall be able to assign an Academic Advisor to each student.

Advisor Oversight	The system shall provide Advisors with a dashboard to view their assigned students and pending registration approval requests.

Table 3.1 User Authentication Requirements

3.1.2 Functional Requirements

3.2 Course Registration & Management

3.2.1 Description

This feature handles the core academic operations, allowing students to search, register, and drop courses while enforcing university rules like prerequisites, credit limits, and timetable conflicts. It also allows Admins to manage course data.

Priority = High.

3.2.2 Functional Requirements

Requirement Name	Details
Course Search	Students shall be able to search for available courses by course name, course code, department, or teacher.
Prerequisite Check	The system shall automatically verify that a student has completed all prerequisite courses before allowing registration. If unmet, the system shall block the registration.
Timetable Conflict	The system shall prevent students from registering for two courses with overlapping class schedules.
Credit Limits	The system shall enforce minimum and maximum credit hour limits for each student's registration.
Registration Window	The system shall allow course registration only during the active registration period set by the Administrator.
Waitlist Queue	If a course reaches maximum capacity, the system shall automatically place the student on a first-come, first-served waitlist.
Course Drop	Students shall be able to drop a registered course anytime before the official Course Withdrawal Deadline. After the deadline, the system shall block the drop.
Course Archiving	The Admin shall be able to archive courses that are no longer offered instead of permanently deleting them, to preserve academic records.

Table 3.2 Course Registration Requirements

3.3 Campus Smart Wallet

3.3.1 Description

This feature provides a unified digital wallet system that allows students to make cashless payments for various campus services (canteen, bus, dorm electricity). It handles wallet top-ups, secure transactions, and balance inquiries.
Priority = Medium.

3.3.2 Functional Requirements

Requirement Name	Details
Wallet Top-up	Students shall be able to add funds to their wallet using an integrated external payment gateway (e.g., credit/debit card).
Pay for Campus Services	Students shall be able to use their wallet balance to pay for cafeteria food, campus bus fares, and dormitory electricity.
Balance & History	Students shall be able to view their real-time wallet balance and a complete transaction history.
Transaction Security	The system shall require a PIN or OTP for wallet transactions that exceed a predefined limit to ensure security.
Insufficient Balance	If the wallet balance is insufficient, the system shall automatically decline the payment and notify the student.
Audit Log	The system shall maintain a tamper-proof audit log of all wallet transactions for administrative review.

Table 3.3 Smart Wallet Requirements

3.4 Exam, Result & Grade Appeal

3.4.1 Description

This feature handles exam schedules, result uploads, automatic CGPA calculation, and the formal grade appeal process to ensure academic fairness.
Priority = High.

3.4.2 Functional Requirements

Requirement Name	Details
Exam Schedule	Students shall be able to view published examination schedules.
Result Upload	Teachers shall be able to upload exam results for their assigned courses.
CGPA Calculation	The system shall automatically calculate and update the student's GPA and CGPA once results are published.
Academic History	Students shall be able to view their academic history and degree progress.
Grade Appeal	Students shall be able to submit a formal grade appeal within a defined period after result publication.
Appeal Tracking	The system shall allow students to track the status of their appeal (e.g., Submitted, Under Review, Approved, Rejected).

Table 3.4 Examination and Result Management Requirements

3.5 Recommendation & Analytics

3.5.1 Description

This feature provides personalized course recommendations based on academic history and generates dashboards/reports for students, advisors, and admins to track progress.
Priority = Medium.

3.5.2 Functional Requirements

Requirement Name	Details
Course Recommendations	The system shall analyze a student's academic history, interests, and degree requirements to recommend suitable courses.
Student Dashboard	The system shall provide a personalized dashboard for students to view registration status, timetable, and recommendations.
Admin & Advisor Reports	The system shall allow Admins and Advisors to generate reports on registration trends and student academic progress.
Registration Patterns	The system shall track registration patterns to improve future course recommendations.

Table 3.5 Recommendation & Analytics

3.6 Campus Life & Facilities

3.6.1 Description

This feature manages non-academic campus services, including library book borrowing, dormitory room applications, and campus event registrations.
Priority = Low.

3.6.2 Functional Requirements

Requirement Name	Details
Library Management	Students shall be able to search for books, view borrowed books, and receive overdue notifications.
Book Reservation	Students shall be able to reserve library books online.
Dormitory Application	Students shall be able to apply for dormitory rooms and view available rooms and accommodation status.
Room Management	The Admin shall be able to manage room allocations and update accommodation status.
Campus Events	Students shall be able to view and register for campus events.
Event Management	Admins shall be able to create, update, and cancel campus events.

Table 3.5 Campus Life & Facilities

3.7 Communication & Notifications

3.7.1 Description

This feature allows users to communicate directly with each other and ensures they receive important automated system updates.
Priority = Medium.

3.7.2 Functional Requirements

Requirement Name	Details
Messaging System	Students shall be able to communicate directly with their assigned Advisors and Teachers via the system.
Announcements	Admins shall be able to post important announcements to all users or specific groups.
System Notifications	The system shall send alerts regarding registration deadlines, approval/rejections, waitlist updates, course changes, and timetable updates.
Mobile Push Alerts	The system shall deliver notifications to students' registered mobile devices.

Table 3.7 Communication & Notifications

4. Data Requirements

4.1 Logical Data Model

The system's data model defines the core entities (Student, Course, Registration, Wallet, Result) and their relationships.

A Student can register for multiple Courses.

- Each Course has multiple Registrations.
- A Student has one Smart Wallet with multiple Transactions.
- A Teacher uploads Results for enrolled students.

4.2 Data Dictionary

The following table defines the key data elements used in the system:

Data Element	Description	Data Type	Length	Allowed Values
Student_ID	Unique identifier for a student	Alphanumeric	12	Numeric only
Course_Code	Unique code for a course	Alphanumeric	8	e.g., CSE-101
Wallet_Balance	Current money in the wallet	Decimal	10,2	0.00 to 999999.99
CGPA	Student's cumulative grade	Decimal	3,2	0.00 to 4.00
Transaction_ID	Unique ID for wallet payment	Alphanumeric	20	Auto-generated

Table 4.2 Data Dictionary

4.3 Reports

The system shall generate the following reports:

- Student Timetable Report: A weekly schedule of registered courses.
- Registration Summary: A list of all students registered in a specific course (for Teachers).

- Wallet Transaction Report: A detailed history of top-ups and payments (for Students and Admins).
- Academic Transcript: A summary of grades and CGPA for a student.

4.4 Data Acquisition, Integrity, Retention, and Disposal

- **Data Acquisition:** Student and course data will be initially imported by the Administrator from the university's central database. Wallet data will be acquired through user input (top-ups).
- **Data Integrity:** The system shall enforce referential integrity (e.g., a student cannot register for a course that does not exist).
- **Data Retention:** Academic records (grades, transcripts) shall be retained permanently. Transaction logs shall be retained for a minimum of 5 years for auditing.
- **Data Disposal:** Obsolete user accounts shall be archived securely. Personal data shall be anonymized upon request, following university privacy policies.

5. External Interface Requirements

5.1 User Interfaces

UI-1: The system shall provide a responsive web interface that is easily accessible and fully functional on both desktop computers and mobile devices (smartphones and tablets).

UI-2: The system shall follow standard accessibility guidelines (e.g., WCAG 2.1 Level AA) to ensure it is usable by people with diverse abilities.

UI-3: The system shall display error messages in clear, non-technical language so that students and faculty can easily understand and resolve issues.

5.2 Software Interfaces

SI-1: Payment Gateway (For Smart Wallet)

- **SI-1.1:** The system shall integrate with an external payment gateway (e.g., Stripe or local banking APIs) to process online wallet top-ups securely.
- **SI-1.2:** The system shall receive payment confirmation status from the gateway before updating the student's wallet balance.

SI-2: University Single Sign-On (SSO)

- **SI-2.1:** The system shall interface with the university's existing SSO system (e.g., Google Workspace or Azure AD) to authenticate users and retrieve their basic profile information (name, email, role).

SI-3: Academic Database

- **SI-3.1:** The system shall interface with the university's central academic database to fetch and validate course prerequisites and student academic history.

5.3 Hardware Interfaces

HI-1: The system shall be accessible via standard input devices such as keyboards and mice for desktop users, and touchscreens for mobile users.

HI-2: If the Smart Wallet supports physical ID cards for canteen payments, the system shall interface with campus card readers to process tap-and-pay transactions.

5.4 Communications Interfaces

CI-1: The system shall use secure HTTPS/TLS protocols for all data communications between the user's browser and the server.

CI-2: The system shall interface with an email/SMS gateway to send automated notifications (e.g., registration deadlines, OTPs for wallet transactions, grade publications).

6. Quality Attributes

6.1 Usability Requirements

- **USE-1:** The system shall provide an intuitive user interface so that a new student can successfully complete course registration without requiring prior training.
- **USE-2:** The system shall allow users to navigate the dashboard and search for courses using a maximum of 3 clicks from the home page.
- **USE-3:** The system shall ensure that 95% of new users can complete their first course registration without errors.

6.2 Performance Requirements

- **PER-1:** The system shall process course registration requests and confirm the seat allocation within 3 seconds under normal load.
- **PER-2:** The system shall support a minimum of 5,000 concurrent users during peak registration periods without significant performance degradation.
- **PER-3:** Web pages generated by the system shall download completely within 2 seconds from the time the user requests the page over a standard 10 Mbps internet connection.

6.3 Security Requirements

- **SEC-1:** All network transactions involving financial information (Smart Wallet) or personally identifiable information (PII) shall be encrypted using TLS/SSL protocols.
- **SEC-2:** The system shall require users to log in via the University SSO before accessing any academic or wallet features.
- **SEC-3:** The system shall store all passwords and PINs using strong hashing algorithms (e.g., bcrypt) and salt.
- **SEC-4:** The system shall lock a user account for 15 minutes after 5 consecutive failed login attempts.
- **SEC-5:** The system shall ensure that students can only view their own academic records and wallet transactions, preventing unauthorized data access.

6.4 Safety Requirements

- **SAF-1:** The system **shall** perform automatic daily backups of all academic and financial data to prevent data loss in case of server failure.
- **SAF-2:** The system **shall** not permanently delete any student academic records or transaction histories; instead, it **shall** archive them securely.

6.5 Other Requirements (Data Consistency & Availability)

- **O-1 (Availability):** The system shall maintain an uptime of 99.9% during the active registration and result publication windows.
- **O-2 (Data Consistency):** The system shall process wallet transactions as "all-or-nothing" (ACID properties). It **shall not** deduct money from a wallet without confirming the service was paid for.

- **O-3 (Reliability):** The system shall ensure zero data loss for course registrations and grade uploads.

Appendix A: Glossary

- **SRS:** Software Requirements Specification.
- **RBAC:** Role-Based Access Control (Users can only access features permitted by their role).
- **MFA:** Multi-Factor Authentication (A security system requiring more than one method to verify identity).
- **SSO:** Single Sign-On (Allows users to log in with one set of credentials across multiple systems).
- **CGPA:** Cumulative Grade Point Average.
- **API:** Application Programming Interface (Allows different software systems to communicate).
- **ACID:** Atomicity, Consistency, Isolation, Durability (Database transaction properties).
- **Smart Wallet:** A digital wallet system used for campus micro-payments.
- **Waitlist:** A queue for students waiting for a seat to open in a full course.